1. Library Management System

1.1 Introduction

Libraries are vibrant hubs of learning and research, balancing thousands of print titles, academic journals, and digital assets alongside a diverse community of readers. Trying to keep track of catalog entries, checkout ledgers, return dates, and overdue penalties with paper logs or spreadsheets is slow, prone to mistakes, and nearly impossible to audit cleanly.

A modern **Library Management System (LMS)** replaces those manual pain points with an intuitive, automated platform. By centralizing cataloging, circulation, reservation queues, and fine collection into a single digital hub, the system eliminates administrative drag for librarians, speeds up checkouts, and gives patrons instant access to search, reserve, and manage their borrowed materials from anywhere.

1.2 Software Requirements Specification (SRS)

1.2.1 Purpose of this Document

This SRS document outlines the functional, technical, and operational specifications for developing the Library Management System. It serves as a single source of truth for software engineers, database administrators, library managers, head librarians, and institutional leadership, ensuring complete alignment across every phase of design, testing, and deployment.

1.2.2 Scope of the System

The system automates the complete operational lifecycle of a library. Key capabilities include catalog acquisition, member registration, circulation workflows (issues, returns, renewals), hold queues, automated overdue fine calculation, barcode/RFID scanning, and collection usage reporting.

- **Estimated Project Cost:** $30,000
- **Target Delivery Timeline:** 5 Months

1.2.3 System Overview

The platform serves as an integrated digital operational workspace linking library staff, patrons, and digital catalog assets. By replacing paper-based desk workflows with automated software services, it keeps inventory records accurate, minimizes lost assets, and speeds up desk checkouts. Key modules include:

- **Online Public Access Catalog (OPAC) & Search**
- **Patron Management & Membership Control**
- **Circulation Engine (Issue, Return & Renewal)**
- **Reservation Queue & Automated Holds**
- **Fine Management & Payment Ledger**
- **Analytics & Administrative Reporting**

1.2.4 General Description

Core Functions

- **Catalog Management:** Create and maintain digital catalog records (ISBN, Title, Author, Publisher, Call Number, Shelf Location) and control physical copy inventories.
- **Patron Administration:** Process member enrollments, issue membership cards, maintain borrowing profiles, and enforce category-based loan rules.
- **Circulation Control:** Execute rapid checkouts, process physical returns, record renewals, and update real-time item availability statuses.
- **Holds & Overdue Tracking:** Manage reservation queues for checked-out materials and trigger automated email/SMS reminders for approaching or past-due return dates.
- **Fines & Revenue Tracking:** Automatically compute daily overdue penalties, process fine payments, and generate digital receipts.

User Objectives

- **Library Staff:** Eliminate manual ledger entries, speed up desk transactions with barcode/RFID scanners, and maintain clear record accuracy.
- **Patrons (Students, Faculty, Public):** Offer self-service search capabilities, easy online book reservations, and hassle-free account renewals.
- **Library Administrators:** Gain actionable visibility into inventory utilization, lost book ratios, peak circulation hours, and popular titles to guide collection acquisition budgets.

System Users

- **Librarians & Circulation Staff:** Desk operators responsible for cataloging new arrivals, managing patron profiles, and handling checkouts/returns.
- **Patrons:** Students, academic faculty, researchers, or general public members who search the collection, borrow items, and manage personal accounts.
- **System Administrators:** IT staff responsible for managing user roles, configuring loan rules, maintaining server health, and overseeing system backups.

Key Benefits

- Significantly faster circulation desk throughput with barcode/RFID integration.
- Improved patron satisfaction through 24/7 online catalog searching and hold requests.
- Reduced loss of library inventory through accurate loan logging and automated due-date reminders.
- Data-backed reporting to optimize book acquisition budgets and track overdue fines.

1.2.5 Functional Requirements

- **Catalog Management**
    - Support full CRUD (Create, Read, Update, Delete) operations for catalog items including ISBN, Title, Author, Publisher, Publication Year, Category, Call Number, and Rack Location.
    - Multi-copy copy tracking (assigning unique physical copy barcode IDs under a single master title ISBN).

    - Real-time search engine supporting keyword queries across Title, Author, Subject, ISBN, or Category with availability filters (**Available / On Loan / Reserved / Under Maintenance**).
- **Patron Management**
    - Register new patrons, issue unique Patron IDs, and maintain detailed profile records.
    - Enforce flexible borrowing policies based on patron category (e.g., *Student*: max 3 books for 14 days; *Faculty*: max 10 books for 30 days).
    - Manage account statuses (**Active, Suspended, Expired, Blocked due to Fines**).
- **Circulation Workflows (Issue, Return, Renewal)**
    - Process checkouts via barcode/RFID scanning, automatically verifying patron eligibility, quota caps, and outstanding fine balances before approval.
    - Process returns, calculate late fees if overdue, and immediately update copy status back to *Available*.
    - Support one-click renewals if no active reservation holds exist on the item.
- **Reservation & Hold Management**
    - Allow patrons to place hold requests on currently checked-out titles via the OPAC portal.
    - Maintain an automated first-in, first-out (FIFO) hold queue per title.
    - Send instant notification alerts (email/SMS) to the reserving patron upon return and place the physical copy on a 48-hour hold shelf.
- **Fine Management & Billing**
    - Automatically calculate daily overdue penalties based on global per-day rate rules.
    - Support manual fine adjustments or waivers by authorized head librarians with required justification notes.
    - Record fine payments (Cash / Online Gateway) and auto-generate digital receipt slips.
- **Administrative Reporting & Analytics**
    - Generate scheduled (daily, weekly, monthly) and custom reports covering: Active Loans, Overdue Items Summary, Unpaid Fine Ledgers, Most Popular/Borrowed Titles, and Inactive Collection Lists.

1.2.6 Interface Requirements

- **Software Interfaces**
    - **Barcode / RFID Peripheral APIs:** Native drivers for fast barcode scanners, RFID readers, and receipt printers at circulation desks.
    - **Messaging Gateways:** SMS and email notification APIs for sending return reminders, hold availability alerts, and fine notices.
    - **Payment Gateway API:** Secure payment aggregator integration for online fine payments via card, net banking, or digital wallets.
    - **Analytics & Data Export:** Data export modules (PDF, CSV, Excel) for operational audits.
- **User Interfaces**
    - **Librarian Workbench:** Clean, high-density desktop web interface optimized for rapid keyboard shortcuts, barcode scanning, and circulation processing.
    - **Patron Portal & OPAC (Mobile-Responsive):** Intuitive self-service catalog search portal allowing patrons to browse titles, place holds, view active loans, and pay fines from mobile or desktop browsers.

1.2.7 Performance Metrics

- **Concurrent Load:** Support up to **500 active concurrent user sessions** without degradation.
- **Search Speed:** OPAC catalog search results render in **under 1.5 seconds** across databases of up to 500,000 records.
- **System Availability:** Ensure **99.5% total operational uptime**.
- **Transaction Error Rate:** Keep transaction processing error rates below **0.05%**.

1.2.8 Technical & Design Constraints

- **Hardware & Infrastructure Constraints**
  - Dedicated server infrastructure with a minimum of 8 GB RAM, 500 GB SSD storage, and an automated off-site backup storage unit.
  - Hardware compatibility with standard USB/Wireless barcode scanners, RFID pads, and thermal receipt printers.
- **Software Constraints**
  - Developed using modern web frameworks (**Java / Spring Boot**, **Python / Django**, or **Node.js**).
  - Relational database backend using **PostgreSQL** or **MySQL** for robust transaction management and data integrity.
- **Standards & Compliance**
  - Adherence to international library cataloging frameworks such as **MARC 21** or **Dublin Core** metadata standards.
  - Full alignment with applicable data privacy regulations regarding patron personal data protection.

1.2.9 Non-Functional Attributes

- **Security:** Role-Based Access Control (RBAC), bcrypt password hashing, encrypted transmission (TLS 1.3), and immutable logs for circulation/fine actions.
- **Portability:** Fully responsive web interface compatible across all modern OS environments (Windows, macOS, Linux, iOS, Android) and browsers.
- **Reliability:** Scheduled nightly automated database backups with quick disaster recovery protocols.
- **Scalability:** Modular database architecture capable of expanding to support multiple library branches, campus locations, and growing catalog collections.
- **Data Integrity:** Foreign-key enforcement and ACID transactional checks to prevent duplicate checkouts or ghost inventory entries.

1.2.10 Project Schedule and Budget

Project Timeline (5 Months Total)

1. **Requirements Gathering & Domain Modeling:** 1 Month
2. **System Architecture, Database Schema & UI/UX Design:** 1 Month
3. **Core Module Development & Hardware Integration:** 2 Months
4. **Testing, QA, Staff Training & Final Deployment:** 1 Month

Budget Allocation

- **Core Application Development:** $18,000
- **Quality Assurance, Hardware Testing & Security:** $6,000
- **Deployment, On-Site Setup & Initial Maintenance:** $6,000
- **Total Estimated Budget: $30,000**
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1.3 Class Diagram:

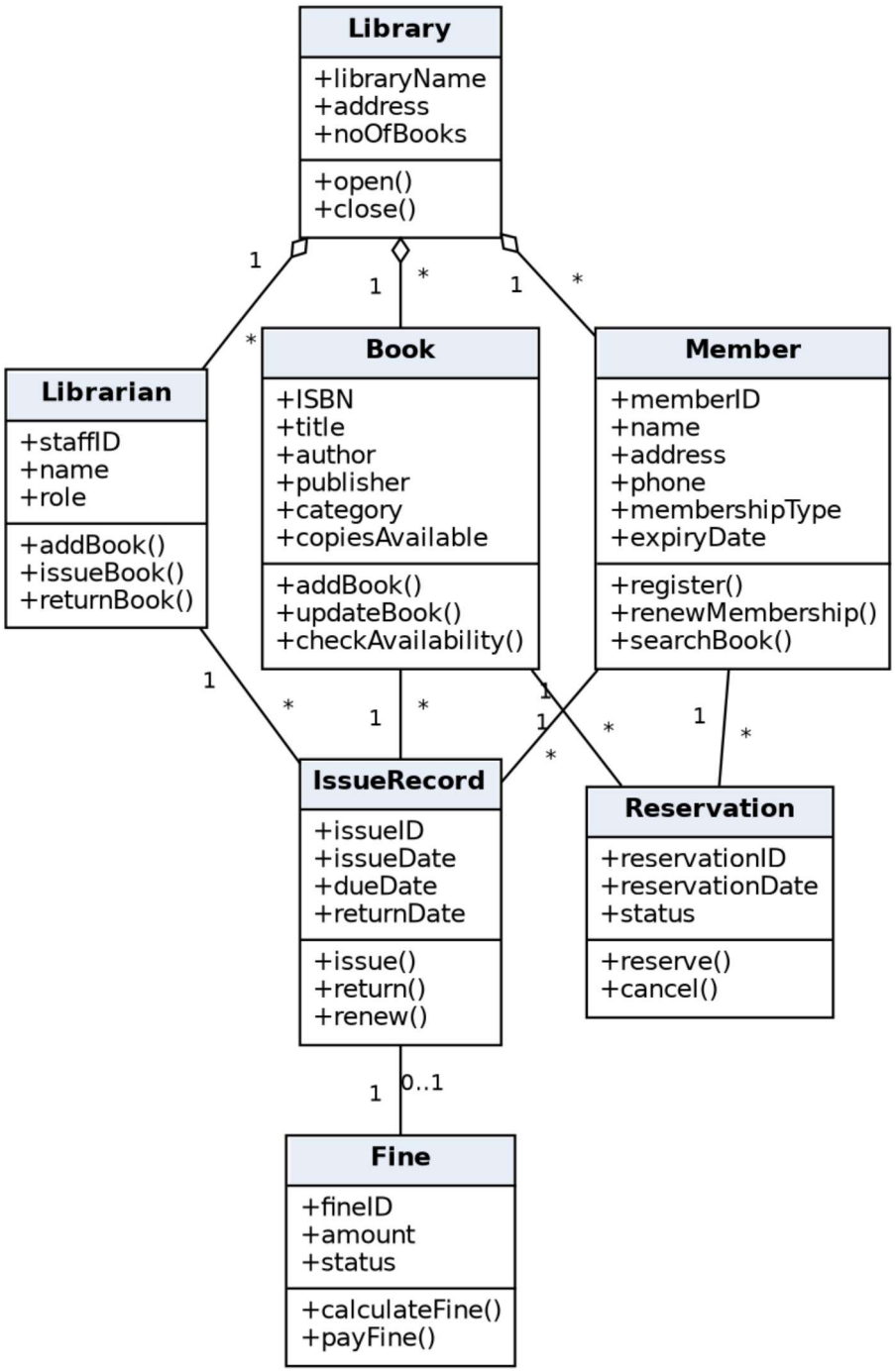


Fig 1.1

Brief Description:

The class diagram for the Library Management System (LMS) represents the key entities, their attributes, and relationships in the system. Below are the main components typically included in the class diagram for LMS:

- **Library:** the main class that aggregates members, books and librarians.

- **Member:** stores member details and membership type; a member can borrow and reserve many books.
- **Librarian:** staff who add books and perform issue and return operations.
- **Book:** holds ISBN, title, author, publisher, category and the number of copies available.
- **IssueRecord:** links a member, a book and a librarian with issue date, due date and return date.
- **Fine:** is generated for an overdue issue record and tracks the amount and payment status.
- **Reservation:** records a member's request to reserve a book that is currently unavailable.